

Adaptive Context-Aware Distillation for Industrial Image Anomaly Detection

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Abstract—Image anomaly detection is extremely challenging in industrial manufacturing processes due to unforeseen and diversified anomalies. Recently, unsupervised anomaly detection methods based on knowledge distillation have been developed and have shown remarkable potential. While most existing methods are devoted to knowledge generalization, they are inadequate for the fine-grained detection task. To address this issue, we propose a novel adaptive context-aware distillation (ACAD) paradigm that gives due consideration to distillation component dependencies and knowledge transfer optimization. Technically, a novel adaptive distillation module (ADM) is proposed for optimal context-aware knowledge transfer, which consists of contrastive decoupling distillation (CDD) and masked perceiving distillation (MPD). The proposed CDD helps to constrain the distribution of different semantic patterns and strengthen the discriminative capability. Vanilla methods treat every pixel as an equal contribution and fail to focus on critical information. To this end, the MPD is proposed to weigh different contextual knowledge adaptively. Extensive experiments with mainstream anomaly detection datasets show that ACAD outperforms the state-of-the-art competitors in accuracy and efficiency. In addition, the experimental results with a real-world inkjet printing organic electroluminescence display (OLED) panel dataset further demonstrate the effectiveness of our method.

Index Terms—Anomaly detection, context-aware optimization, contrastive learning, knowledge distillation.

I. INTRODUCTION

ANOMALY detection is vital to guarantee product quality in complex industrial manufacturing processes. The primary goal of anomaly detection is to detect outliers and locate anomalies that deviate from normal instances. These anomalies generally manifest as local structure damages such as scratches, dents, contaminations, or logical violations such as permissible objects occurring in invalid locations [1]. Currently, efficient and reliable visual inspection technology has been widely applied to product quality control instead of traditional manual inspection approaches. However, considering the

tremendous and unforeseeable variations of surface anomalies, accurate anomaly detection is still a challenging but integral part of real-world industrial applications.

In the past decade, the rapid development of deep neural networks has significantly advanced the anomaly detection task with powerful representation capabilities [2]. Particularly, unsupervised methods can be applied to cases where it is difficult to collect sufficient abnormal samples. Most early methods used reconstruction-based architectures [3], [4], [5], [6], [7], [8], [9], [10] that detect outliers under the assumption that inliers are significantly more frequent and share a common concept of normalcy [11]. These methods can produce images with per-pixel generation, such as typical autoencoders (AEs) and generative adversarial networks (GANs). The reconstruction residuals, derived from the disparity between an original image and its reconstructed counterpart, are used to distinguish anomalies. However, these methods are prone to bias toward reconstructing abnormal regions well and obtaining incorrect per-pixel estimation.

Recently, increasing studies based on embedding similarity have been devoted to anomaly detection by exploring the potential of rich feature manifolds [12], [13], [14], [15], [16], [17]. These methods perform well in extracting meaningful embedding features and project raw images into a high-dimensional feature space, where anomalies are much more distinguishable. Typically, [14] and [15] use the Gaussian distribution to construct a correlation model of normal training samples. Roth et al. [17] adopted an extensive memory bank to store and leverage available normal patch features optimally. Since the normal distribution in the feature space is constrained in fixed regions, the features distributed in the remaining regions are identified as anomalies. Thus, anomaly scores can be derived by calculating the distances between embedding vectors of test and train samples. However, these methods cannot meet the real-time requirement in practical applications due to the massive computations of vector-level similarity distances and considerable memory consumption for storing patch embeddings.

In addition to the methods mentioned above, unsupervised methods based on knowledge distillation [18], [19], [20], [21], [22], [23], [24], [25], [26] have recently been developed. These methods attempt to transfer meaningful knowledge from other well-studied computer vision tasks pretrained with natural datasets. Generally, these methods train a student network that learns to mimic the output of a pretrained teacher network and detect anomalies through their output disparities. Inchoate

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methods tended to adopt identical or similar architectures for both student and teacher networks to improve the efficiency of knowledge transfer. Then, the student attempts to learn the normal pattern distribution during training and tries to generalize on previously unseen data [27]. Therefore, during testing, the student's output for the learned distribution closely resembles that of the teacher. For test data outside the normal distribution, the output from the student significantly differs from that of the teacher, resulting in different anomaly scores. The pixel with a more significant anomaly score indicates a higher likelihood of being abnormal. However, existing methods utilize a homogeneous distillation paradigm, which cannot fully exploit the context-aware information, leading to weak discriminative capabilities in the inference process.

To address the abovementioned issues, we propose a novel adaptive context-aware distillation (ACAD) paradigm in this study. Compared to previous work, ACAD transfers optimal knowledge with distinct architectures and data flows to enhance the diversity of pattern representations. It is difficult for convolutional neural networks (CNNs) to learn a precise decision boundary from only normal samples. Generally, CNNs tend to overgeneralize on normal samples and perform anomaly localization poorly. Thus, we introduce artificial anomalies to mitigate the offset inconsistency between the training and testing data distributions. Based on the assumption that normal and abnormal patterns can be decoupled from abnormal samples, ACAD leverages the semantic class-separability for normality reconversion and anomaly segmentation. To achieve superior performance, a novel adaptive distillation module (ADM) is proposed for knowledge transfer optimization. This module encourages the network to focus more on valuable knowledge while considering contextual information dependencies. At the inference stage, our method has the potential to detect and localize various anomalies precisely through the joint detection of the teacher and student networks. Extensive experiments and analyses conducted with the mainstream anomaly detection dataset MVTec AD [28] and a real-world inkjet printing OLED panel dataset demonstrate that ACAD achieves the state-of-the-art performance without using real-world abnormal samples. The main contributions of this study are summarized as follows.

- 1) A novel ACAD paradigm is proposed for anomaly detection and achieves the new state-of-the-art performance. ACAD adopts heterogeneous distillation architectures and considers dependencies of different distillation components with vital context-aware optimization.
- 2) A novel contrastive decoupling distillation (CDD) loss is proposed to facilitate the network to learn discriminative representations of different semantic patterns. CDD utilizes the embeddings projected into the latent space by the teacher network as prior knowledge to explicitly pull normal features closer together while pushing outliers further apart to boost superior performance.
- 3) A novel masked perceiving distillation (MPD) approach is proposed to weigh different contextual knowledge for normal pattern reconversion adaptively. This innovation introduces a set of trainable tokens to customize

weighted masks, which help to focus on critical information and refine the perception capacity.

The remainder of this article is organized as follows: In Section II, previous works related to our method are briefly reviewed. In Section III, the proposed ACAD is illustrated in detail. In Section IV, the extensive experiments and analyses conducted to evaluate our method are presented. Finally, we conclude this article and discuss future studies in Section V.

II. RELATED WORKS

In this section, for a more intuitive comparison, we summarize previous representative works along with their advantages and limitations in a tabular format according to categories, as shown in Table I. In addition to the reconstruction-based methods and embedding similarity-based methods introduced in Section I, we briefly review previous research contributions related to our work in Sections II-A–II-C.

A. Anomaly Simulation-Based Methods

Since it is impracticable to collect and annotate sufficient abnormal samples, it is expected that models can generalize to deal with unforeseen and diversified types of anomalies while only having access to normal samples during training. Anomaly simulation-based methods attempt to generate abundant pseudo-abnormal samples with pixel-level annotations to alleviate the sample imbalance problem. Schlüter et al. [37] proposed an intuitive self-supervision method named natural synthetic anomalies (NSAs). NSA integrates Poisson image editing to blend scaled patches of various sizes from separate images seamlessly and creates abundant synthetic anomalies that are more similar to natural irregularities. Li et al. [38] proposed a simple data augmentation strategy named CutPaste that cuts a rectangular region of a certain image and pastes it at a random location in another image. CutPaste performs well in image-level anomaly detection, but its anomaly localization precision is limited. For increased diversity, Zavrtanik et al. [29] proposed a discriminatively trained reconstruction anomaly embedding model (DRAEM) to generate anomalies based on Perlin noise and auxiliary datasets. In addition, a pair of reconstructive and discriminative subnetworks was utilized to produce high-fidelity per-pixel anomaly detection maps. Liu et al. [32] proposed a simple and application-friendly network called SimpleNet that generates anomaly features by adding Gaussian noise to the adapted normal features. Then, a simple discriminator consisting of a few multi-layer perceptron (MLP) layers was trained on these features to discriminate anomalies. Yang et al. [31] proposed a multcategory decomposition editing network (MCDEN) that adopts a simulative defect generation strategy similar to CutMix to make the model more sensitive to matching normal and abnormal areas. MCDEN uses a defect library with abnormal data to recognize and decompose defect regions into assumed fundamental conversions. Similarly, Song et al. [30] proposed an anomaly composition and decomposition network (ACDN). ACDN introduces a Gaussian-sampling-based anomaly composition (GSAC) method to composite a large number of images that are similar to real defective

TABLE I
SUMMARY OF THE ADVANTAGES AND LIMITATIONS OF PREVIOUS REPRESENTATIVE ANOMALY DETECTION METHODS

Category	Representative Methods	Advantages	Limitations
Reconstruction	AE-SSIM(2019) [4], AFEAN(2020) [7], OCRGAN(2022) [9], MMR(2023) [10]	Typically unsupervised and conceptually simple and intuitive.	Easy to reconstruct anomalies as well and obtain incorrect per-pixel estimation.
Embedding similarity	SPADE(2020) [12], Patch-SVDD(2020) [13], PaDiM(2021) [15], PatchCore(2022) [17]	Capable of capturing high-level patterns and complex information.	Sensitive to the quality and quantity of training data. Generally need extra storage and time overhead.
Anomaly simulation	DRAEM(2021) [29], ACDN(2022) [30], MCDEN(2023) [31], SimpleNet(2023) [32]	Alleviate the need for extensive data collection and labeling efforts in real-world industrial scenarios.	Quality and diversity of synthetic anomalies may not always perfectly match real-world anomalies.
Semantic segmentation	AnoSeg(2021) [33],SGSF(2022) [34], SIDN(2022) [35], JRCC-Net(2023) [36]	Understand contextual relationships and provide fine-grained pixel-level anomaly segmentation results.	Demand extensive labeled data. Potential difficulties in handling low contrast and subtle anomalies.
Knowledge distillation	US(2020) [18], STPM(2021) [20], IKD(2022) [22], RD4AD(2022) [24]	Transfer knowledge from other well-studied pretrained networks and improve generalization and efficiency.	May neglect knowledge distillation component dependencies and fail to obtain fine-grained detection results.

images for training. Due to the adoption of the GAN structure, GSAC is prone to mode collapse during training.

B. Semantic Segmentation-Based Methods

The semantic segmentation task is an important processing step for scene understanding or better explaining the global context of an image [39]. Previously, U-Net [40], a typical semantic segmentation-based network with a U-shape structure to obtain contextual and positional information, was proposed to solve the problem of medical image segmentation and achieved outstanding performance. Due to the excellent pixel-level classification performance, increasing segmentation models have been proposed to address anomaly detection tasks. Xing et al. [34] proposed a novel self-supervised guided segmentation framework (SGSF) for anomaly detection by jointly exploring the effective generation method of forged anomalous samples and the normal sample features as the guidance information of segmentation. Zhang et al. [36] proposed a segmentation network with joint representation and contrast clustering (JRCC-Net), which is able to locate anomalies directly in an end-to-end manner. Song et al. [33] proposed an anomaly segmentation network called AnoSeg that generates detailed anomaly maps using holistic approaches to maximize segmentation performance while requiring generative reconstruction for the entire input image. Without involving reconstruction, Yang et al. [35] proposed a semantic information decomposition network (SIDN) for efficient texture defect segmentation by combining texture and defect segmentation results. However, SIDN cannot guarantee optimal performance for textures with irregular patterns, such as tile and wood in [28].

C. Knowledge Distillation-Based Methods

Knowledge distillation was originally generalized by Hinton et al. [41] as a means of model compression and knowledge transfer. The core idea of knowledge distillation is to transfer meaningful knowledge from a cumbersome teacher

to a smaller and faster student. It is widely acknowledged that the teacher model's dark knowledge provides privileged information on the similarity of class categories to enhance student learning [42]. Recently, some methods based on knowledge distilled from pretrained networks have become potential solutions to anomaly detection. Bergmann et al. [18] proposed an uninformed students (US) framework to train an ensemble of student networks with anomaly-free training data to mimic the output of a pretrained teacher network. During testing, the output disparity between the teacher and student networks is coupled with the uncertainty among student predictions to localize anomalies. In contrast to US, Wang et al. [20] proposed a student-teacher feature pyramid matching (STPM) network to enable a student network to receive a mixture of multilevel knowledge from a feature pyramid. STPM and US share a similar method of knowledge transfer, but STPM distills the knowledge of hierarchical matching features into a single student network with an identical architecture. Following STPM, Cao et al. [22] proposed an information knowledge distillation (IKD) method, which transfers informative knowledge to alleviate the overfitting problem with a strong supervisory signal. IKD has achieved promising performance on object category anomaly detection but failed on texture categories. These methods adopt similar or identical architectures to construct the teacher-student (T-S) framework, inhibiting anomalous representation diversities. To further enhance the discriminating capability on various anomalies, Deng and Li [24] proposed a framework named anomaly detection via reserve distillation (RD4AD) to use an encoder and a decoder to construct a reverse distillation paradigm. Nevertheless, the student network in RD4AD still receives inputs from the teacher network, which might readily leak abnormal information to the student network during testing, failing to generate discrepant features on anomalies.

III. METHOD

In this section, the proposed method is introduced in detail. First, the overall architecture is briefly depicted. Then, the

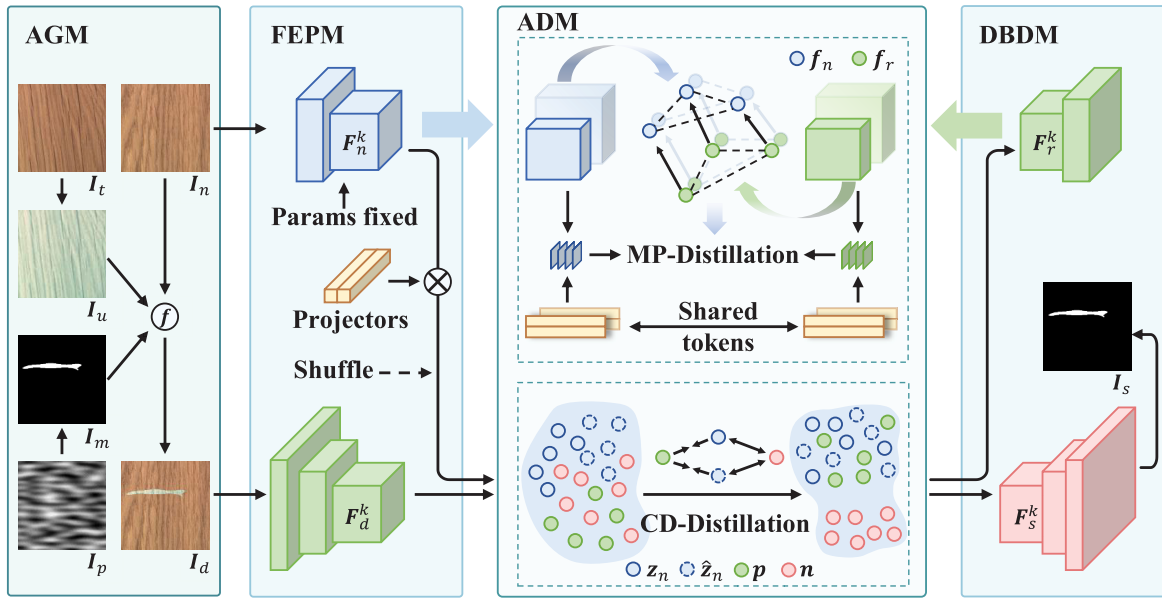


Fig. 1. Illustration of the proposed ACAD. ACAD consists of four components: an AGM, a FEPM, an ADM, and a DBDM.

main components of ACAD are presented in detail. Finally, the training and testing procedures are described.

A. Architecture of ACAD

The overall architecture of ACAD is illustrated in Fig. 1. Specifically, an anomaly generation module (AGM) is first employed to generate artificially pseudo-abnormal samples. As a result, it provides an explicit mechanism to enable the model to distinguish the potential deviations between normal and abnormal samples. Then, both samples are fed into a feature encoding projection module (FEPM), where the two kinds of raw data are encoded by a pretrained teacher network and a randomly initialized student network, respectively. During training, the weight parameters of the teacher network are frozen. To further facilitate the discriminative representations, a novel CDD loss is proposed to constrain normal features to cluster and move away from anomalies. Subsequently, different semantic elements are utilized by a dual branch decoding module (DBDM) for normal pattern reconversion and abnormal region segmentation. The former receives multilevel knowledge and aims to mimic the intermediate embedding features of the teacher network for restoring pseudo-normal patterns with dense descriptive features. The latter understands global semantic information and aims to generate fine-grained anomaly segmentation results. A novel MPD approach is proposed to fully exploit context-aware optimization and valuable knowledge transfer. At the inference stage, the output disparities derived from the T-S and anomaly segmentation results are aggregated together for detection. Extensive experimental results presented in Section IV further demonstrate the significance of each individual component.

B. Anomaly Generation Module

Most existing unsupervised methods only use anomaly-free samples for training, restricting the discriminative capability

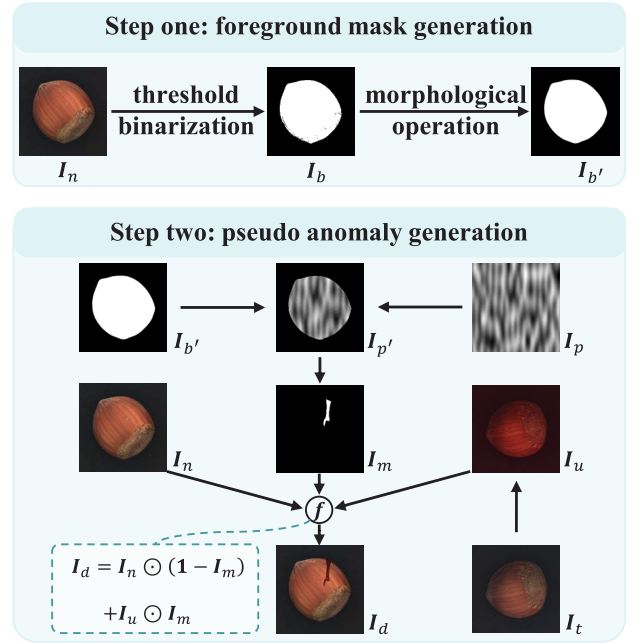


Fig. 2. Illustration of the anomaly generation process for foreground objects.

of the model. To alleviate this problem, we design a simple but effective strategy to generate pseudo-abnormal samples. First, a noise image I_p is generated by a Perlin noise generator, and I_p is binarized by a threshold operation to obtain a binary image that contains diversely irregular shapes. The largest one among them is reserved for generating the mask image I_m . It is worth noting that mask images are generated slightly differently for datasets involving foreground objects. Since random anomaly masks tend to appear in the background and cause unnecessary interference, the foreground mask of an object is obtained in advance through threshold binarization and morphological operations, as shown in Fig. 2. Then, anomaly-free images I_t and I_n are randomly sampled from

the original training dataset using non-replacement random sampling, accompanied by the flipping or rotating operation. Inspired by DRAEM [29] and RandAugment [43], three kinds of augmented transformations randomly selected from the set $\{\text{gamma-contrast, brightness change, hue change, solarize, invert, GaussianBlur, elastic transformation}\}$ are successively applied to I_t to generate the augmented image I_u . The augmented transformations in the set belong to an image augmentation package (called imgaug) in Python. Finally, the pseudo-abnormal image I_d is composited by fusing I_n with I_u and I_m

$$I_d = I_n \odot (1 - I_m) + I_u \odot I_m \quad (1)$$

where $I_d, I_n, I_u \in \mathbb{R}^{3 \times H \times W}$, and $I_m \in \mathbb{R}^{1 \times H \times W}$. Here, 3 or 1, H and W denote the channel, height, and width of the images, respectively. $\odot$ denotes the element-wise multiplication operation, namely, the Hadamard product.

C. Feature Encoding Projection Module

To fully exploit the semantic and structural information of the input data, the FEPM is introduced to perform feature extraction and projection procedures. In this study, the first two residual blocks of ResNet [44] are chosen as teacher feature extractors, which are frozen during training. The normal sample I_n is forwarded into a powerful teacher pretrained with the ImageNet [45] classification dataset to extract hierarchical features. Then, the corresponding pseudo-abnormal sample I_d is handled by a designed student encoder, which consists of three basic convolution blocks. Each block contains two convolution layers with kernel sizes of 3 and 3 and strides of 2 and 1. The number of channels in the first block is 32, and the number of channels in the subsequent two blocks gradually increases by a multiple of 2. Unless otherwise specified, all convolution or transposed convolution layers are followed by ReLU and BatchNorm layers in the remainder of this article. Only the last output feature of the student encoder layer is used for subsequent tasks. Thus, the feature extraction procedure can be formulated as follows:

$$\{\mathbf{F}_n^k\}_{k=1}^2 = f_{\text{Te}}(I_n; \theta_{\text{Te}}) \quad (2)$$

$$\{\mathbf{F}_d^k\}_{k=1}^3 = f_{\text{Se}}(I_d; \theta_{\text{Se}}) \quad (3)$$

where $\mathbf{F}_n^k, \mathbf{F}_d^k \in \mathbb{R}^{C_k \times H_k \times W_k}$ denote hierarchical feature maps. $f_{\text{Te}}(\cdot)$ and θ_{Te} denote the function and parameters of the teacher encoder, respectively. $f_{\text{Se}}(\cdot)$ and θ_{Se} denote the function and parameters of the student encoder, respectively.

Furthermore, considering that the teacher network is derived by training with the natural dataset ImageNet, there is generally a data domain gap between the source and target datasets. Hence, a set of projectors is introduced to fine-tune the feature $\mathbf{F}_n^2$ and map its vector representation of each spatial position into a new latent space $\{\mathbf{z}_n(x, y)\}_{x=0, y=0}^{H_l, W_l}$, which constitutes the embedding $\mathbf{Z}_n$

$$\mathbf{Z}_n = f_{\text{pt}}(\mathbf{F}_n^2; \theta_{\text{pt}}) \quad (4)$$

where $\mathbf{Z}_n \in \mathbb{R}^{C_l \times H_l \times W_l}$. $f_{\text{pt}}(\cdot)$ and θ_{pt} denote the function and parameters of the projectors, respectively. To preserve the

original space dimension while projecting each embedding position to an independently distributed vector, two stacked 1×1 convolution layers are employed to form the projectors. Note that these layers will be discarded at the inference stage without introducing extra costs.

D. Contrastive Decoupling Distillation

Due to the tremendous variations in surface anomalies, previous works have been unable to sufficiently distinguish the differences in the aspect of feature representations, especially for subtle and low-contrast anomalies. Contrastive learning [46], a technique typically employed in self-supervised representation learning, has shown its efficacy in learning discriminative representations by comparing positive and negative pairs. Inspired by this, a novel CDD loss is proposed to maximize the divergence among different semantic features.

As mentioned in Section III-C, the embeddings of the teacher network are projected into the latent space as prior knowledge to guide the student network. Similarly, $\mathbf{F}_d^3$ can be viewed as a set of local feature vectors $\{\mathbf{z}_d(x, y)\}_{x=0, y=0}^{H_l, W_l}$, which constitutes the embedding $\mathbf{Z}_d$ in the spatial dimension to align with $\mathbf{Z}_n$. To construct positive and negative pairs for the contrastive decoupling procedure, the pixel-level class label map $\mathbf{I}_l \in \mathbb{R}^{1 \times H_l \times W_l}$ is obtained by downsampling from the ground truth to classify different semantic vectors. Because anomaly detection can be simplified as a binary classification task in which normalities are unified and other outliers are regarded as anomalies, we define the sets of normal and abnormal semantic embedding vectors as $\{\mathbf{p}_i\}_{i=1}^P$ and $\{\mathbf{n}_j\}_{j=1}^N$, respectively,

$$\{\mathbf{p}_i\}_{i=1}^P = \{\mathbf{z}_d(x, y) | \mathbf{I}_l(x, y) = 0\} \quad (5)$$

$$\{\mathbf{n}_j\}_{j=1}^N = \{\mathbf{z}_d(x, y) | 1 - \mathbf{I}_l(x, y) = 0\} \quad (6)$$

where $i = 1, \dots, P$, and $j = 1, \dots, N$. P and N denote the numbers of elements in the respective sets. In addition, $x = 1, \dots, H_l$, and $y = 1, \dots, W_l$. For ease of notation, the cosine similarity function between two vectors, such as $\mathbf{v}_i$ and $\mathbf{v}_j$, is defined as $\text{sim}(\mathbf{v}_i, \mathbf{v}_j)$ that is formulated as follows:

$$\text{sim}(\mathbf{v}_i, \mathbf{v}_j) = \frac{(\mathbf{v}_i)^T \cdot \mathbf{v}_j}{\|\mathbf{v}_i\|_2 \|\mathbf{v}_j\|_2} \quad (7)$$

where $\|\cdot\|_2$ denotes the $\mathcal{L}_2$ -norm.

In this study, the proposed CDD loss is executed in two parts. A similarity comparison of embedding vectors in the same spatial position is first developed between $\mathbf{Z}_d$ and $\mathbf{Z}_n$ without other interferences. In this case, $\mathbf{Z}_d$ and $\mathbf{Z}_n$ are strictly aligned in the batch dimension, that is, the normal pixel information of $\mathbf{Z}_d$ and $\mathbf{Z}_n$ is derived from the same image. Consequently, referring to the popular InfoNCE [47] loss, a pixel-level contrastive loss L_{intra} is devised to pull the normal inliers close and push abnormal outliers apart

$$L_{\text{intra}} = \frac{1}{P} \sum_{i=1}^P \log \left(1 + \frac{\sum_{j=1}^N \exp(\text{sim}(\mathbf{n}_j, \mathbf{z}_n)/\tau)}{\exp(\text{sim}(\mathbf{p}_i, \mathbf{z}_n)/\tau)} \right) \quad (8)$$

where $\mathbf{z}_n$ denotes the vector belonging to $\{\mathbf{z}_n(x, y)\}_{x=0, y=0}^{H_l, W_l}$ and has the same spatial position as $\mathbf{p}_i$ or $\mathbf{n}_j$. $\tau > 0$ denotes a temperature hyperparameter.

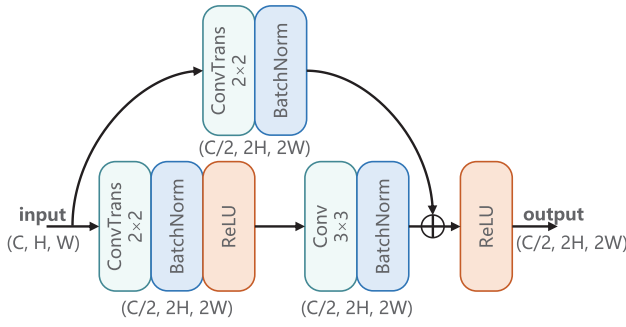


Fig. 3. Implementation details of the reverse residual structure.

Considering the high similarity of normalities among a batch of training samples, the distribution of normalities from different samples should also be homogeneous. To further promote learning robust representations, we introduce a perturbation to $\mathbf{Z}_n$ through random batch shuffling and obtain a batch non-aligned set $\{\hat{\mathbf{z}}_n(x, y)\}_{x=0, y=0}^{H_i, W_i}$, which constitutes the embedding $\hat{\mathbf{Z}}_n$. With the same form as L_{intra} , a cross-image contrastive loss L_{inter} is then devised to constrain the feature distribution across global samples

$$L_{\text{inter}} = \frac{1}{P} \sum_{i=1}^P \log \left(1 + \frac{\sum_{j=1}^N \exp(\text{sim}(\mathbf{n}_j, \hat{\mathbf{z}}_n)/\tau)}{\exp(\text{sim}(\mathbf{p}_i, \hat{\mathbf{z}}_n)/\tau)} \right). \quad (9)$$

Finally, the proposed novel CDD loss, denoted as L_{cdd} , can be formulated as follows:

$$L_{\text{cdd}} = \frac{1}{2} (L_{\text{intra}} + L_{\text{inter}}). \quad (10)$$

Through the CDD, the class-separability of pixel-level features between normalities and anomalies is significantly enhanced, which boosts the discriminative capability of the model.

E. Dual Branch Decoding Module

As shown in Fig. 1, given the embedding features preliminarily decoupled to involve respective category-specific information, the DBDM is introduced to further decode and retain favorable information in each task branch.

1) *Normality Reconversion*: Following [24], the normality reconversion branch is designed to be symmetrical but reverse compared to the teacher and produces matching hierarchical features. It is worth noting that the inputs of the branch are not derived from the teacher network, which clearly distinguishes our approach from previous methods. Specifically, corresponding to the first two residual blocks of ResNet, there are two reverse residual blocks designed for decoding. The detailed decoding structure is depicted in Fig. 3. The outputs of the branch are formally denoted as follows:

$$\{\mathbf{F}_r^k\}_{k=1}^2 = f_{Sr}(\mathbf{Z}_d; \theta_{Sr}) \quad (11)$$

where $\mathbf{F}_r^k \in \mathbb{R}^{C_k \times H_k \times W_k}$ denotes the recovered feature. $f_{Sr}(\cdot)$ and θ_{Sr} denote the function and parameters of the branch, respectively. Note that $\mathbf{F}_r^k$ has the same shape as $\mathbf{F}_n^k$.

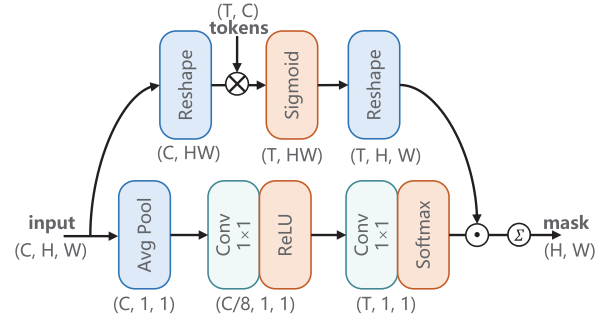


Fig. 4. Implementation details of the perceptual mask.

2) *Anomaly Segmentation*: Knowledge distillation-based methods are generally insufficient in terms to guarantee fine-grained detection results with clear boundaries when inspecting subtle anomalies, and our normality reconversion branch is no exception. Therefore, an anomaly segmentation branch is introduced in our method to produce auxiliary segmentation results for joint detection. The decoding structure of the segmentation branch is symmetrical to that of the student encoder, as shown in Fig. 1. This structure consists of three basic transposed convolution blocks and an output layer. Every basic transposed convolution block is designed by replacing the first convolution layer with a pair of 2×2 transposed convolution layer and skip connection. The final output layer of the branch is a 2×2 transposed convolution followed by a sigmoid activation layer. The anomaly segmentation procedure for obtaining the image $\mathbf{I}_s$ is formulated as follows:

$$\mathbf{I}_s = f_{Ss}(\mathbf{Z}_d; \theta_{Ss}) \quad (12)$$

where $\mathbf{I}_s \in \mathbb{R}^{1 \times H \times W}$ denotes the anomaly segmentation image. $f_{Ss}(\cdot)$ and θ_{Ss} denote the function and parameters of the branch, respectively. Then, the per-pixel cross-entropy loss is used as the anomaly segmentation loss

$$L_{\text{seg}} = \mathbb{E}_{\mathbf{I}_d} [-\|\mathbf{I}_m \odot \log \mathbf{I}_s + (1 - \mathbf{I}_m) \odot \log(1 - \mathbf{I}_s)\|_1] \quad (13)$$

where $\mathbb{E}[\cdot]$, $\odot$, and $\|\cdot\|_1$ denote the expectation calculation, Hadamard product, and $\mathcal{L}_1$ -norm, respectively.

F. Masked Perceiving Distillation

Existing knowledge distillation-based methods are generally dedicated to generalizing the expert knowledge and treating every pixel as an equal contribution to knowledge transfer. This motivation neglects different knowledge properties and restricts performance. Inspired by [48] and [49], to adaptively emphasize or suppress contextual knowledge transfer, a novel MPD approach is proposed to focus on critical information and refine the perception capacity.

As shown in Fig. 4, the trainable embedding tokens $\mathbf{E}$ are introduced to generate masks containing pixels of interest

$$\mathbf{M}_n^k = f_{mg}(\mathbf{F}_n^k, \mathbf{E}^k; \theta_{mg}) \quad (14)$$

$$\mathbf{M}_r^k = f_{mg}(\mathbf{F}_r^k, \mathbf{E}^k; \theta_{mg}) \quad (15)$$

$$\mathbf{M}^k(x, y) = \mathbf{M}_n^k(x, y) \odot \mathbf{M}_r^k(x, y) \quad (16)$$

where $\mathbf{M}_n^k, \mathbf{M}_r^k, \mathbf{M}^k \in \mathbb{R}^{H_k \times W_k}$, and $\mathbf{M}^k$ denotes the learned holistic pixel-level mask. $\mathbf{E}^k \in \mathbb{R}^{T \times C_k}$, and T denotes the number of underlying semantic primaries that deserve to be captured. $f_{\text{mg}}(\cdot)$ and θ_{mg} denote the function and parameters of the mask generation procedure, respectively. Then, $\mathbf{F}_r^k(x, y)$ and $\mathbf{F}_n^k(x, y) \in \mathbb{R}^{1 \times H_k \times W_k}$ are used to represent feature vectors at position (x, y) in each feature map, respectively. Dense pixel-level supervision is adopted to make the normality reconversion branch mimic the outputs of the teacher network, similar to conventional [20]. First, the $\mathcal{L}_2$ -distance along the channel dimension between the normalized feature vectors is formulated as follows:

$$\mathbf{D}^k(x, y) = \left\| \frac{\mathbf{F}_r^k(x, y)}{\|\mathbf{F}_r^k(x, y)\|_2} - \frac{\mathbf{F}_n^k(x, y)}{\|\mathbf{F}_n^k(x, y)\|_2} \right\|_2^2 \quad (17)$$

where $\mathbf{D}^k \in \mathbb{R}^{H_k \times W_k}$, and $x = 1, \dots, H_k, y = 1, \dots, W_k$. Then, the k th hierarchy distillation loss for pixel-level knowledge transfer can be formulated as follows:

$$L_k = \frac{1}{2H_k W_k} \sum_{x=1}^{H_k} \sum_{y=1}^{W_k} ((1 + \mathbf{M}^k(x, y)) \odot \mathbf{D}^k(x, y)) \quad (18)$$

where the learned mask is given equal confidence to superimpose on the conventional pixel-level regression. Finally, the loss for the entire MPD process can be formulated as follows:

$$L_{\text{mpd}} = \sum_{k=1}^K L_k \quad (19)$$

where K denotes the number of feature hierarchies used for the experiment in this study.

G. Adaptive Distillation Module

Considering the dependencies between distillation components, the ADM is proposed for optimal context-aware knowledge transfer, which essentially introduces CDD and MPD separately during the encoding and decoding phases to facilitate the learning of optimal representations. During training, the final outputs of CDD and MPD are composed of corresponding distillation loss. Hence, ADM can be summarized and formulated as follows:

$$\begin{aligned} L_{\text{adm}} &= \lambda_1 f_{\text{cdd}}(\mathbf{Z}_d, \mathbf{Z}_n, \hat{\mathbf{Z}}_n; \theta_{\text{cdd}}) \\ &\quad + \lambda_2 f_{\text{mpd}}(\{\mathbf{F}_r^k, \mathbf{F}_n^k, \mathbf{E}^k\}_{k=1}^K; \theta_{\text{mpd}}) \\ &= \lambda_1 L_{\text{cdd}} + \lambda_2 L_{\text{mpd}} \end{aligned} \quad (20)$$

where λ_1 and λ_2 are tunable hyperparameters to balance loss terms. $f_{\text{cdd}}(\cdot)$ and θ_{cdd} denote the function and parameters of CDD, respectively. $f_{\text{mpd}}(\cdot)$ and θ_{mpd} denote the function and parameters of MPD, respectively.

H. Training and Inference Procedure

During training, the overall architecture of ACAD is optimized by the joint loss function

$$L = \alpha L_{\text{adm}} + \beta L_{\text{seg}} \quad (21)$$

where α and β are tunable hyperparameters to balance loss terms. Using (21), repeating optimization steps to continuously

Inference Procedure

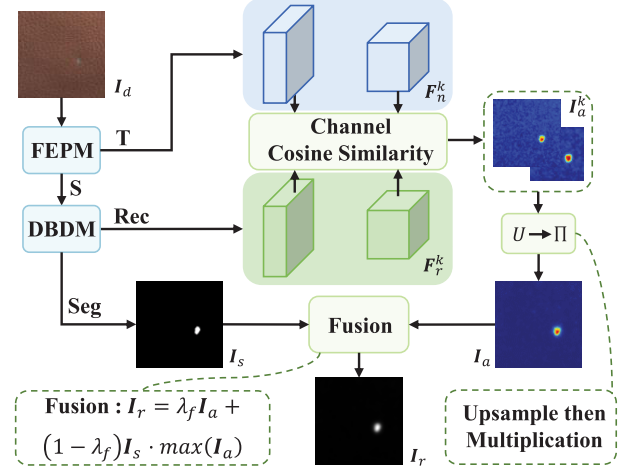


Fig. 5. Inference procedure of our method. The abbreviations T and S represent the feature data extracted by the teacher and student networks, respectively. The abbreviations Seg and Rec represent the anomaly segmentation and the normality reconversion, respectively.

update the parameters ensures that the network converges after reaching the specified number of epochs.

In the inference procedure as shown in Fig. 5, given a test image $\mathbf{I}_d \in \mathbb{R}^{3 \times H \times W}$, multiscale anomaly maps are first derived from the output disparities between the normality reconversion branch ($\mathbf{F}_r^k$) and the pretrained teacher ($\mathbf{F}_n^k$). Referring to (7), we use the cosine similarity to calculate hierarchical anomaly maps

$$\mathbf{I}_a^k(x, y) = 1 - \text{sim}(\mathbf{F}_r^k(x, y), \mathbf{F}_n^k(x, y)) \quad (22)$$

where $\mathbf{I}_a^k \in \mathbb{R}^{1 \times H_k \times W_k}$. Moreover, the anomaly maps first need to be upsampled to the same size as $\mathbf{I}_d$ by bilinear interpolation. Then, taking the contributions of hierarchical upsampled anomaly maps into equal consideration, the element-wise multiplication operation is adopted to obtain the holistic anomaly map

$$\mathbf{I}_a = \prod_{k=1}^K U(\mathbf{I}_a^k) \quad (23)$$

where $\mathbf{I}_a \in \mathbb{R}^{1 \times H \times W}$, and U denotes the upsampling operation. Finally, $\mathbf{I}_a$ is aggregated together with the anomaly segmentation image $\mathbf{I}_s$ to obtain the detection result $\mathbf{I}_r$

$$\mathbf{I}_r = \lambda_f \mathbf{I}_a + (1 - \lambda_f) \mathbf{I}_s \cdot \max(\mathbf{I}_a) \quad (24)$$

where $0 \leq \lambda_f \leq 1$ is used to adjust the weights of two terms and $\max(\mathbf{I}_a)$ denotes the maximum value in $\mathbf{I}_a$.

IV. EXPERIMENTS

In this section, a series of experiments are conducted to qualitatively and quantitatively evaluate the performance of ACAD and its individual components. Additionally, ACAD is applied to inspect inkjet printing OLED panels to verify its effectiveness in practical applications.

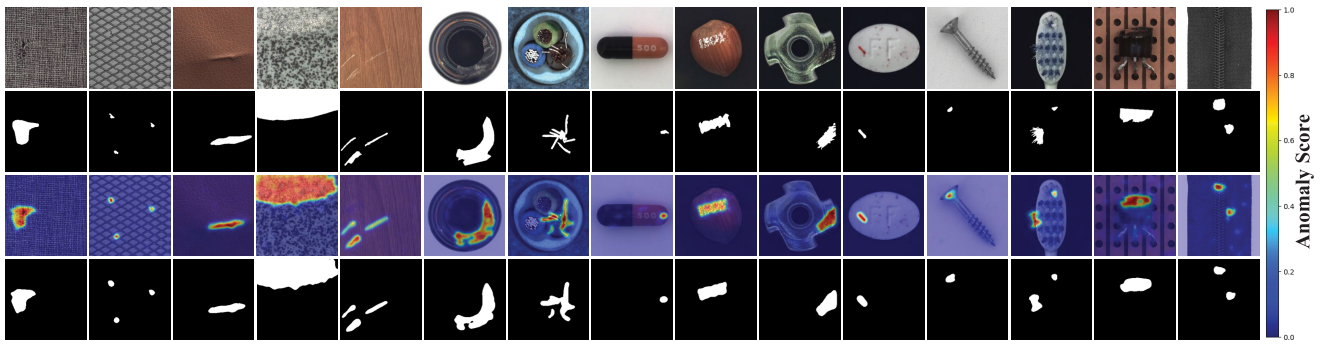


Fig. 6. Visualization of some qualitative detection results on the MVTec AD dataset [28]. Rows from top to bottom correspond to test images, ground truth images, detection results after Gaussian filtering, and images obtained through a threshold operation.

A. Dataset and Evaluation Criteria

1) *MVTec AD Dataset*: The MVTec anomaly detection (MVTec AD) [28] is a challenging dataset to benchmark surface anomaly detection and localization algorithms. It contains a series of industrial abnormal and normal images, including five categories of textures and ten categories of objects. There are a total of 3629 normal images for training and a mixture of 1725 normal and abnormal images for testing. All images are high-resolution images of real-world industrial products, most of which are well center-aligned. Moreover, abnormal images of each category contain diverse anomalies with pixel-level ground truth annotations for evaluation.

2) *DAGM Dataset*: The DAGM dataset [50] contains ten categories of various textures with artificially generated anomalies, which are extremely subtle and closely resemble the texture background, making it particularly challenging for unsupervised anomaly detection. The dataset comprises a total of 8050 training images and 8050 testing images, some of which contain anomalies and have been roughly labeled.

3) *Evaluation Criteria*: The pixel-level area under the receiver operating characteristic (AuROC) [28] and the normalized area under the per-region-overlap (AuPRO) [18] are adopted for evaluation on the MVTec AD dataset. For the AuROC and AuPRO metrics, a higher evaluation score indicates better detection performance. The AuROC is a standard anomaly detection metric computed from the receiver operating characteristic (ROC) curve, which is a curve of the true-positive rate (TPR) with respect to the false-positive rate (FPR) under different thresholds. In contrast to the AuROC, the AuPRO weights ground-truth regions of different sizes equally. To calculate the AuPRO metric, the anomaly score map needs to be processed by a threshold operation. Then, the overlap percentage of each connected region within the ground truth image with the corresponding region within the anomaly score map is calculated to evaluate performance. Consistent with [18], the AuPRO is calculated with an average per-pixel FPR threshold of 0.3. For evaluation on the DAGM dataset, due to the coarse ground truth labels, it is challenging to measure the anomaly localization performance accurately. Thus, three different evaluation criteria, image-level AuROC, TPR, and true-negative rate (TNR), are used.

B. Implementation Details

During the experiments, the first two residual blocks (i.e., conv2_x and conv3_x) of the ResNet18 [44]

pretrained with ImageNet [45] are selected as the teacher, and its parameters are frozen. The data augmentation strategy in III-B is employed to generate 1280 pseudo-abnormal samples for training, and all samples are resized to the resolution of 256×256 . Specifically, the augmentation operations are set as follows: $\{\text{gamma-contrast}(0.5, 1.5), \text{brightness change}(0.8, 1.2), \text{hue change}(-60, 60), \text{solarize}(32, 128), \text{invert}(\text{default}), \text{GaussianBlur}(\text{default}), \text{elastic transformation}(2, 9)\}$.

For the hyperparameters in (20) and (21), λ_1 , λ_2 , α , and β are set to 0.1, 100, 1, and 10, respectively. In addition, the distillation temperature hyperparameter τ in (8) and (9) is experientially set to 0.1. Because underlying semantic primaries in the anomaly detection task can be roughly abstracted into normalities and anomalies, the T dimension in the trainable embedding tokens E is set to 2. The Adam optimizer [51] is used to train our model with a learning rate of 0.01, and the β is the default value of (0.5, 0.999). The entire training procedure is carried out with a batch size of 32 and 200 iterations. Furthermore, the CosineAnnealingLR strategy [52] is adopted to adjust the learning rate at the end of each epoch, with a minimum value of 0.0001 in the last epoch. At the inference stage, the hyperparameter λ_f in (24) is set to 0.5, which indicates that the final detection result gives equal confidence to the dual decoding branches. After obtaining the detection result using (24), a Gaussian filter with $\sigma = 4$ is used to perform the postprocessing. All the experiments are conducted on a device with an Intel Xeon¹ @2.20 GHz CPU and an NVIDIA Titan Xp GPU, using PyTorch 1.9.0.

C. Comparison With Existing Methods

1) *Experiment on MVTec AD*: In this section, we compare our ACAD with other state-of-the-art methods on the MVTec AD dataset [28] to holistically evaluate the detection performance. Some qualitative detection results of our method are visualized in Fig. 6. Table II reports the quantitative pixel-level AuROC and AuPRO results of our method for anomaly localization compared to those of existing methods: SPADE [12], PaDiM-WResNet50 [15], DRAEM [29], PatchCore [17], STPM [20], IKD [22], RD4AD [24], and MMR [10]. For the sake of fairness and impartiality, the AuROC and AuPRO results of other methods compared in the experiment are taken from the corresponding official papers [10], [12], [15], [17],

¹Registered trademark.

TABLE II

AUROC(%) / AuPRO(%) FOR ANOMALY LOCALIZATION ON THE MVTEC AD DATASET [28]. WE COMPARE OUR METHOD WITH EXISTING SOTA METHODS TO HOLISTICALLY EVALUATE THE PERFORMANCE. FOR BOTH AUROC AND AuPRO, THE BOLD FONT INDICATES THE BEST RESULT

Category/Method		SPADE [12]	PaDiM [15]	DRAEM [29]	PatchCore [17]	STPM [20]	IKD [22]	RD4AD [24]	MMR [10]	Ours
Textures	Carpet	97.50/94.70	99.10/96.20	95.50/-	99.10/96.60	98.80/95.80	98.71/94.49	98.90/97.00	98.80/96.60	99.58/98.10
	Grid	93.70/86.70	97.30/94.60	99.70/-	98.70/95.90	99.00/96.60	97.04/87.73	99.30/ 97.60	99.00/96.50	99.40/97.54
	Leather	97.60/97.20	99.20/97.80	98.60/-	99.30/98.90	99.30/98.00	98.53/97.64	99.40/99.10	99.20/98.60	99.70/99.22
	Tile	87.40/75.90	94.10/86.00	99.20/-	95.90/87.40	97.40/92.10	95.68/86.35	95.60/90.60	95.60/90.20	99.28/95.67
	Wood	88.50/87.40	94.90/91.10	96.40/-	95.10/89.60	97.20/93.60	93.88/89.06	95.30/90.90	94.80/88.90	98.27/95.86
	Average	92.94/88.38	96.90/93.20	97.90/-	97.62/93.68	98.34/95.22	96.77/91.05	97.70/95.00	97.48/94.16	99.25/97.28
Objects	Bottle	98.40/95.50	98.30/94.80	99.10/-	98.60/96.10	98.80/95.10	98.99/96.08	98.70/ 96.60	98.30/96.00	99.03/96.49
	Cable	97.20/90.90	96.70/88.80	94.70/-	98.50/92.60	95.50/87.70	98.03/ 94.21	97.40/91.00	95.40/87.20	96.70/90.36
	Capsule	99.00/93.70	98.50/93.50	94.30/-	98.90/95.50	98.30/92.20	98.55/90.62	98.70/ 95.80	98.00/94.50	98.00/95.43
	Hazelnut	99.10/95.40	98.20/92.60	99.70/-	98.70/93.90	98.50/94.30	98.71/95.97	98.90/95.50	98.50/91.20	98.93/ 97.45
	Metal Nut	98.10/94.40	97.20/85.60	99.50/-	98.40/91.30	97.60/94.50	98.38/ 94.69	97.30/92.30	95.90/88.60	95.55/94.66
	Pill	96.50/94.60	95.70/92.70	97.60/-	97.60/94.10	97.80/96.50	98.79/96.09	98.20/96.40	98.40/96.10	98.92/97.89
	Screw	98.90/96.00	98.50/94.40	97.60/-	99.40/97.90	98.30/93.00	98.63/92.95	99.60/98.20	99.50/97.60	99.30/96.75
	Toothbrush	97.90/93.50	98.80/93.10	98.10/-	98.70/91.40	98.90/92.20	98.58/87.01	99.10/94.50	98.40/93.00	99.23/95.08
	Transistor	94.10/87.40	97.50/84.50	90.90/-	96.40/83.50	82.50/69.50	97.13/ 93.78	92.50/78.00	90.20/79.10	87.56/88.93
	Zipper	96.50/92.60	98.50/95.90	98.80/-	98.90/97.10	98.50/95.20	97.56/91.55	98.20/95.40	98.00/95.00	98.76/96.57
	Average	97.57/93.40	97.80/91.60	97.00/-	98.41/93.34	96.47/91.02	98.34/93.30	97.90/93.40	97.06/91.83	97.20/ 94.96
Total Average		96.03/91.73	97.50/92.11	97.30/-	98.14/93.45	97.09/92.42	97.81/92.55	97.80/93.90	97.20/92.61	97.88/ 95.73

[20], [22], [24], [29]. The AuROC and AuPRO metrics of our method presented in Table II are obtained by calculating the mean result of three repeated training experiments.

As Table II displays, our method significantly outperforms all existing state-of-the-art methods in terms of both AuROC and AuPRO metrics on texture categories. Our method achieves the best average outcome, with a **99.25%** AuROC score and a **97.28%** AuPRO score. It is worth noting that none of the existing methods can simultaneously reach high accuracy on five texture categories except ours. Additionally, our method achieves the maximum average AuPRO scores of **94.96%** and **95.73%** on object categories and overall 15 categories, indicating the superiority of our method. With a competitive average AuROC score of **97.88%** across all categories, our method is only 0.26% points lower than the 98.14% score of PatchCore. For the average AuROC score on texture categories, our method surpasses the second-best STPM by a large margin of 0.91% points. Furthermore, our method surpasses the second-best RD4AD by a large margin of 1.83% points for the overall AuPRO average score. Most existing methods perform poorly in particular categories (i.e., wood and toothbrush) due to their relatively irregular and subtle anomalies. In contrast, our method makes notable improvements on these categories.

In particular, although STPM and IKD also employ the knowledge distillation strategy and the T-S framework, they still fail to achieve satisfactory results. The main reason is that the knowledge transfer in their student and teacher networks is homogeneous, which hinders their discriminative capabilities in the inference process. Besides, RD4AD adopts the reverse design to facilitate activation discrepancies on anomalies,

but its performance declines substantially in some intricate categories (i.e., tile and wood). Although RD4AD attempts to avoid a similar structure and knowledge generalization during distillation, its student network still receives inputs from the teacher network, which readily leaks abnormal information to the student network and leads to inferior prediction in inference. Moreover, all these methods neglect to reinforce the discrepant representations and context-aware knowledge optimization in the feature space. They merely introduce the output of one network into another network as usual.

However, although our method holistically achieves excellent performance across 15 categories, there are still some shortcomings. Regarding texture categories, our method achieves the second-best detection performance on the grid category with only a slight gap from the best method. Regarding object categories, our method has significant advantages in terms of the AuPRO metric but does not perform well in evaluating the AuROC metric. Specifically, our method achieves the best detection performance on the pill and toothbrush categories. Our method obtains competitive results on the bottle and zipper categories and achieves the second-best performance. Although the AuPRO score of our method on the hazelnut category is the highest, the corresponding AuROC score is slightly inferior to that of DRAEM [29], which is sensitive to large-scale discriminative segmentation. Compared to SPADE [12] and RD4AD [24], which use deep pretrained network features and contain a large number of parameters, our method achieves inferior performance on the capsule and screw categories. The reason may be that our method uses only shallow pretrained network features, prompting our method to be lightweight and efficient but limiting the capability of the

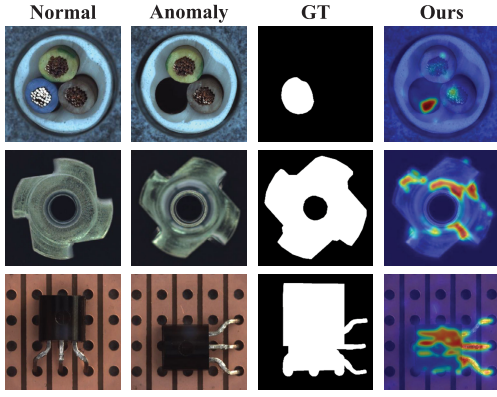


Fig. 7. Some examples of anomalies involving violations of logical constraints in the categories of cable (missing cable), metal nut (flip), and transistor (misplaced) [28]. Columns from left to right correspond to normal images, abnormal images, ground-truth images and failed detection results.

TABLE III

COMPARISON OF THE DETECTION PERFORMANCE ON THE DAGM DATASET [50]. THE BOLD INDICATES THE BEST RESULTS OF THE UNSUPERVISED (UNSUP) AND SUPERVISED (SUP) METHODS

	Method	AuROC(%)	TPR(%)	TNR(%)
Unsup.	US [18]	72.50	72.60	65.30
	PaDiM [15]	95.00	83.30	97.50
	DRAEM [29]	99.00	96.50	99.40
	JRCC-Net [36]	97.20	-	-
	Ours	99.25	97.82	99.60
Sup.	Racki <i>et al.</i> [53]	99.60	99.85	99.46
	Lin <i>et al.</i> [54]	99.00	99.45	99.93
	Božič <i>et al.</i> [55]	100	100	100

network to capture deep semantic information. In addition, our method fails to achieve satisfactory performance on the cable, metal nut and transistor categories. Among these three categories, there are some anomalies involving violations of logical constraints, as shown in Fig. 7. PaDiM [15] and PatchCore [17] achieve excellent results in detecting these anomalies due to their use of the memory bank mechanism, which may restrict the inference efficiency of the model. It can be inferred that our method lacks an explicit mechanism to constantly encourage the network to store and update normal patterns during training.

2) *Experiment on DAGM*: In experiments on the DAGM dataset [50], only anomaly-free samples from the training set are utilized for training, following the methodology of the previous experiment. Since the anomalies are only coarsely labeled, the detection performance is evaluated by classifying whether the test image contains anomalies rather than measuring the accuracy of anomaly localization.

The quantitative results of unsupervised methods shown in Table III are obtained from [29] and [36], and the quantitative results of supervised methods are obtained from the corresponding official papers [53], [54], [55]. As shown in Table III, our method surpasses other state-of-the-art unsupervised methods and remains competitive compared to supervised methods. Furthermore, although there is still a slight gap between our

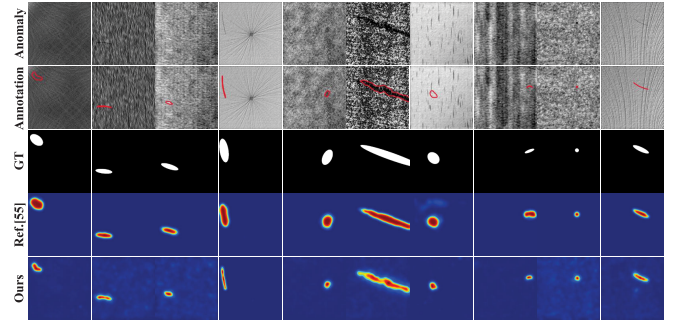


Fig. 8. Comparison of some visualized results on the DAGM dataset [50]. Compared to the best supervised method [55], our method achieves more accurate anomaly localization without using labeled training samples.

TABLE IV

COMPARISON OF PARAMETER SIZE AND INFERENCE SPEED ON THE MVTEC AD DATASET [28] WITH 256×256 RESOLUTION INPUTS

Method	PaDiM [15]	PatchCore [17]	STPM [20]	RD4AD [24]	Ours
Size(MB)	3,891.20	131.82	89.32	483.89	50.94
Speed(FPS)	1.05	4.55	9.62	3.48	27.68

method and the best supervised method [55], our method achieves more accurate anomaly localization, as shown in Fig. 8. Because the training labels of supervised methods are roughly annotated with ellipses that approximately cover the anomalies, their anomaly localization results are not sufficiently accurate. In contrast, our method achieves precise anomaly localization without using labeled training samples.

3) *Complexity Analysis*: To further analyze the complexity of ACAD, we compare it with other methods from two aspects: parameter size and inference speed, and the quantitatively specific results are shown in Table IV. Compared with the embedding similarity-based methods, such as PaDiM [15] (with the backbone of WResNet50) and PatchCore [17], our method not only contains much fewer parameters but also surpasses them in terms of inference speed by a large margin. Since PaDiM and PatchCore need to store an enormous number of features and calculate anomaly scores with a complex feature search and matching process, they introduce considerable extra memory and time costs. In particular, the time consumption of PaDiM and PatchCore increases linearly with the input size due to the patch-level computation during inference. In contrast to knowledge distillation-based STPM [20] and RD4AD [24], our method still has apparent advantages in terms of efficiency and storage resource consumption, which benefit the potential for practical applications. With a lightweight pretrained backbone, STPM achieves comparable storage resource consumption to ours, whereas its detection accuracy and inference speed are significantly worse.

D. Ablation Analysis of ACAD

A series of ablation experiments are analyzed in this section to further investigate the effectiveness of individual components of the proposed ACAD. All ablation experiments are

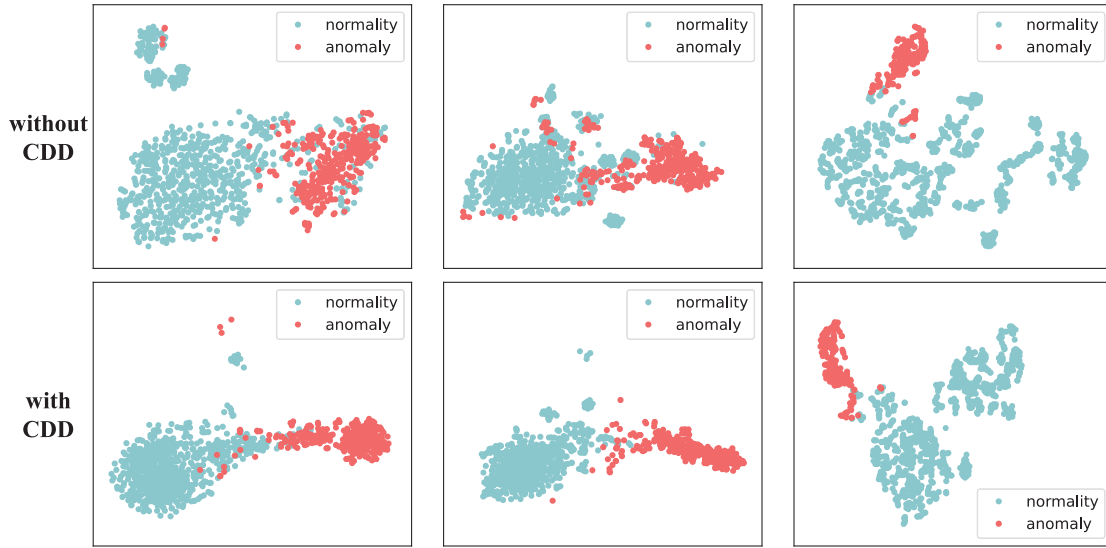


Fig. 9. Some visualization examples of learned feature vectors from ACAD without (top row) or with (bottom row) CDD by using t-SNE [56]. Columns from left to right correspond to the test samples selected from the tile, wood, and bottle categories.

TABLE V
ABLATION STUDY FOR EFFECTS OF AGM, CDD, AND MPD. THE ABSENCE OF MPD IS REPLACED BY THE VANILLA MEANS INTRODUCED IN [20]

AGM	L_{intra}	L_{inter}	MPD	AuROC(%) / AuPRO(%)
				96.34 / 93.19
✓				97.21 / 95.17
✓	✓			97.37 / 95.29
✓		✓		97.38 / 95.26
✓	✓	✓		97.47 / 95.51
✓			✓	97.64 / 95.45
✓	✓	✓	✓	97.88 / 95.73

conducted with the MVTec AD dataset [28] using the AuROC and AuPRO criteria.

1) *Effectiveness of AGM*: Since it is impracticable to collect and annotate sufficient abnormal samples in real-world industrial environments, AGM is introduced as a means of data augmentation to generate artificial anomalies. To investigate the effects of AGM, we conducted experiments both with and without it. When AGM is not employed, the model is unable to utilize CDD, MPD, and the segmentation branch of DBDM, because it can only be trained with normal samples. In particular, the absence of MPD is replaced by the vanilla means introduced in [20]. The quantitative effects are reported in Table V, where the performance of the model exhibits a significant improvement with the influence of AGM. Comparing the first two rows of Table V, the average AuROC and AuPRO scores increase by 0.87% and 1.98%, respectively. Our method is based on the assumption that normal and abnormal patterns can be decoupled from abnormal samples. Therefore, without AGM, our method essentially degrades into training the reconversion of normal patterns solely using normal samples. Furthermore, AGM can mitigate the offset inconsistency between the training and testing data distributions.

2) *Effectiveness of CDD*: CDD is proposed to constrain the distribution of different semantic patterns and enhance the discriminative capability. Technically, CDD utilizes the embeddings projected into the latent space by the teacher network as prior knowledge to help the student network to cluster normal features and push anomalies away explicitly. To analyze the effects of CDD more intuitively, we visualize the feature vectors encoded from abnormal samples of the testing dataset by using t-SNE [56], as shown in Fig. 9. The samples are selected from the tile, wood, and bottle categories and forwarded into the network trained with or without CDD, respectively. Obviously, although normal and abnormal feature vectors can be roughly separated without CDD, they are highly scattered with much overlap in the spatial distribution. In contrast, CDD makes the congeneric feature vectors more compact and distinguishable in the feature space and presents a clear boundary between normal and abnormal feature vectors, which visually demonstrates the effectiveness of CDD for promoting the class-separability of different semantic information.

As shown in Table V, in the case of investigating the effects of CDD that consists of L_{intra} and L_{inter} , different loss combinations are adopted in the experiments. The experimental results demonstrate that our method achieves similar improvements in both AuROC and AuPRO when using only L_{intra} or L_{inter} . With the integrated CDD, our method improves the AuROC score by 0.26% and the AuPRO score by 0.34% compared to the basic architecture, which is ACAD without CDD and MPD. Obviously, L_{intra} and L_{inter} are consistent and complementary in promoting the model to distinguish anomalies during training.

The optimization of hyperparameters is a common topic in many tasks [57], [58], [59], and it significantly impacts the overall performance of the model. To explore the influence of temperature parameter τ in (8) and (9), we attempted to train and evaluate our method with different orders of magnitude τ , as shown in Fig. 10. Under the influence of different τ values, both the overall AuROC and AuPRO scores

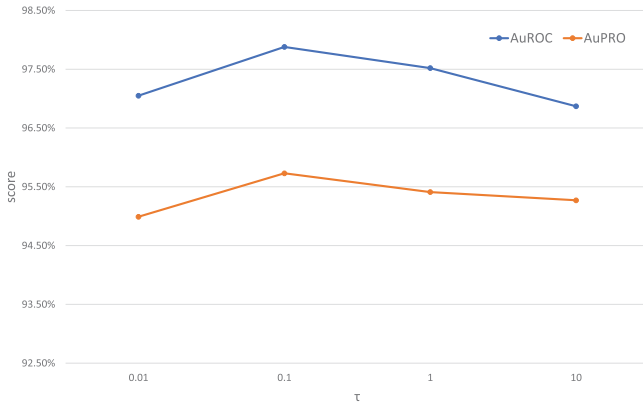


Fig. 10. Line chart of AuROC and AuPRO metrics evaluated with different temperature hyperparameters τ .

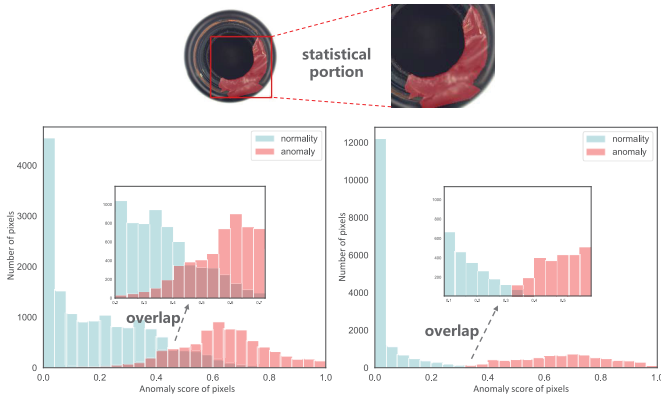


Fig. 11. Histogram of the anomaly score distribution of the detection result image derived without (left) or with (right) MPD. The statistics for anomaly scores come from a portion of the image.

show a trend of first increasing and then decreasing. Due to the properties of exponential functions, the temperature parameter τ is mainly used to adjust the smoothness of the data distribution after exponential operation. However, abnormal feature data is usually much less abundant than normal feature data, which may cause the model not to be fully optimized during training. It can be inferred that when τ is equal to 0.1, the contributions of the two types of feature data can be appropriately regulated.

3) *Effectiveness of MPD*: MPD is proposed to adaptively optimize knowledge transfer and guide the student network to focus more on valuable knowledge with consideration of contextual information dependencies. The ablation experimental results are also presented in Table V, where it is shown that the additional MPD further improves the performance. Obviously, the average AuROC and AuPRO scores are improved by 0.53% and 0.28%, respectively, compared to the basic architecture, which is ACAD without CDD and MPD. In addition, an intuitive comparison example is shown in Fig. 11, where the statistical histogram of the anomaly score distribution is used to quantify the effectiveness of MPD. A portion of the test sample is extracted to calculate the histogram distributions to balance the gap between the number of normal and abnormal pixels for an intelligible visual perspective. Ideally, the anomaly scores of normal pixels should be concentrated around 0, while the anomaly scores of

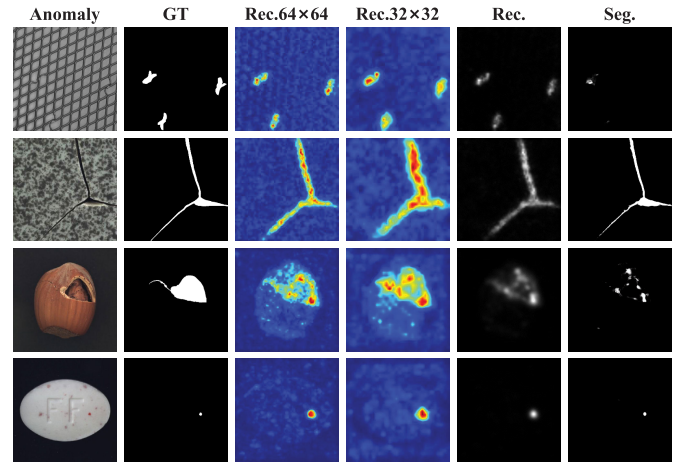


Fig. 12. Detailed detection results of DBDM on the grid, tile, hazelnut, and pill categories. Columns from left to right correspond to test images, ground truth images, anomaly maps of two hierarchies, fused anomaly maps without colormap operations, and anomaly segmentation images.

TABLE VI
ABLATION STUDY FOR EFFECTS OF THE HIERARCHICAL NORMALITY RECONVERSION (REC) AND THE ANOMALY SEGMENTATION (SEG)

Rec.64 × 64	Rec.32 × 32	Seg.	AuROC(%) / AuPRO(%)
✓			96.86 / 94.74
	✓		96.35 / 94.36
✓	✓		97.00 / 94.93
		✓	96.58 / 94.66
✓	✓	✓	97.88 / 95.73

abnormal pixels should be concentrated around 1. Obviously, without MPD, there is a significant overlap between the two different distributions of anomaly scores. In contrast, the overlap significantly decreases when using MPD, and less overlap means less false detection. Thus, the nonoverlapping distribution of normality and anomaly indicates the strong discriminative capability of our model.

4) *Effectiveness of DBDM*: We compare the situations with different decoding combinations to explore the influence of dual task branches, namely the normality reconversion (Rec) and the anomaly segmentation (Seg). Table VI reports the comparison results of evaluation scores, and some detailed visualization results are presented in Fig. 12. It can be seen in Table VI that the performance drops dramatically when either branch is used alone. Both the AuROC and AuPRO average scores reach the highest values only when the detection results of dual branches are used simultaneously. As can be observed in the last two columns of Fig. 12, the anomaly maps produced by the normality reconversion branch are generally rougher but more robust than the anomaly segmentation results, which have more precise boundaries close to the ground truth regions but tend to miss some inconspicuous anomalies or internal anomalies of large-scale structural breakdown, such as bent (grid) and crack (hazelnut). In addition, hierarchical color maps presented in the third and fourth columns of Fig. 12 show different characteristics: low-level features produce high-resolution anomaly maps, which capture small receptive fields; high-level features produce

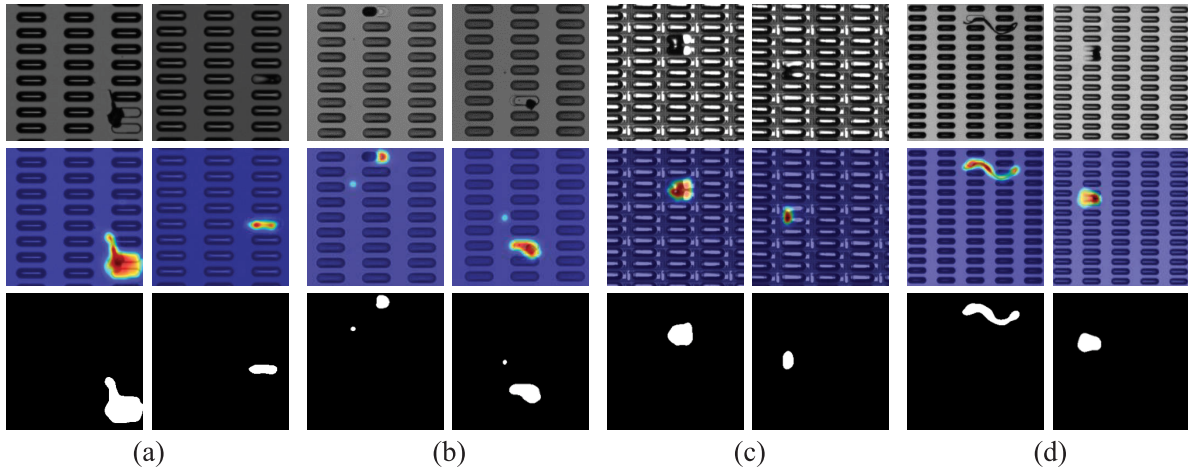


Fig. 13. Visualization of some qualitative detection results on the inkjet printing dataset. Rows from top to bottom correspond to test images, anomaly maps after Gaussian filtering, and results obtained through a threshold operation. Columns from left to right correspond to the images in different detection conditions. (a) Large ink volume. (b) Small ink volume. (c) Tight layout. (d) Low magnification.

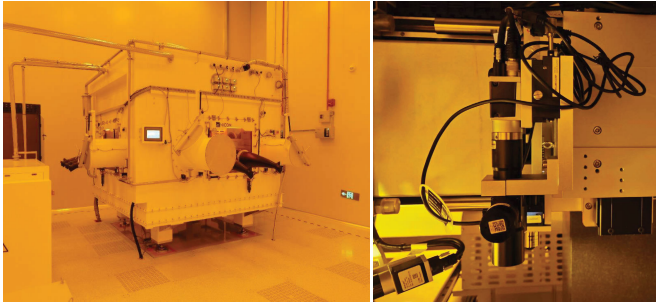


Fig. 14. Inkjet printing equipment (left) and vision inspection system (right).

low-resolution anomaly maps, which carry higher semantic level information and are suitable for detecting large-scale anomalies. Therefore, the aggregation of hierarchical anomaly maps and the segmentation image reflects a trade-off between robustness and fineness, achieving the best comprehensive performance.

E. Inkjet Printing Panels Inspection Application

In this section, experiments on a real-world inkjet printing organic electroluminescence display (OLED) panel dataset are conducted to further evaluate the practical performance of our ACAD. Some visualized experimental results are shown in Fig. 13. The inkjet printing equipment and its automated optical inspection (AOI) system are shown in Fig. 14. Inkjet printing technology dissolves organic molecular materials in specific solvents to form printing ink. Then, it uses a printhead to squeeze the ink into droplets to fill pixel pits. After vacuum drying and curing, a series of substrate layers are formed. The layers are repeatedly printed, cured, and finally packaged to form the finished display device. Due to the complex production processes, various defects are easily produced in each link of inkjet printing manufacturing. These defects seriously restrict the mass production application of inkjet printing technology in large-scale display substrate manufacturing. Therefore, it is essential to monitor the quality of display substrate products.

TABLE VII

DETAILED PRESENTATION OF NUMBER OF IMAGES IN EACH CATEGORY OF THE INKJET PRINTING DATASET

Category/State	Train (normal)	Test (normal)	Test (defective)
Large ink volume	522	65	408
Small ink volume	669	166	672
Tight layout	522	173	58
Low magnification	387	84	528
<i>Total Number</i>	2100	488	1666

TABLE VIII

DETECTION RESULTS ON THE INKJET PRINTING DATASET

Category/Criterion	Precision(%)	Recall(%)	F1 score(%)
Large ink volume	99.75	99.02	99.38
Small ink volume	99.85	99.70	99.78
Tight layout	98.31	100	99.15
Low magnification	98.81	99.24	99.53
<i>Total Average</i>	99.43	99.49	99.46

The inkjet printing dataset was collected from the vision inspection system of the inkjet printing equipment. It contains a total of 2100 defect-free images for training and a mixture of 2154 defect-free and defective images for testing. All images have a resolution of 256×256 . The dataset contains four categories of images: images printed with a large ink volume, images printed with a small ink volume, images printed in a tight layout, and images collected by a low magnification lens. The detailed presentation of number of images in each category is displayed in Table VII. A number of qualitative detection results are visualized in Fig. 13, where various defects on the inkjet printing panels can be detected effectively. To meet the requirements of practical production, we adopt precision, recall, and F1 score as three criteria to evaluate the performance of our method comprehensively. As shown in Table VIII, our method achieves average scores of 99.43%,

99.49%, and 99.46% on the three criteria, respectively. Moreover, our method can achieve a detection speed of about 60 fps for a panel resolution of 256×256 . The experimental results with the inkjet printing dataset further demonstrate the effectiveness of our method.

V. CONCLUSION

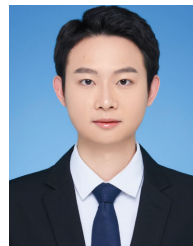
In this study, we propose a novel ACAD paradigm for accurate industrial image anomaly detection. By fully considering distillation component dependencies and significant knowledge transfer optimization, ACAD effectively improves the discriminative capacity in the inference process. Technically, a novel CDD is introduced to enhance the class-separability between normalities and anomalies. Furthermore, a novel MPD is proposed to help focus on critical information and refine the perception capacity, boosting superior performance. The experimental results with mainstream anomaly detection datasets and the inkjet printing dataset demonstrate that ACAD significantly surpasses the existing state-of-the-art methods and achieves high-precision and real-time performance.

However, there remains some future work worth investigating. On the one hand, the proposed method is not good at detecting some anomalies involving violations of logical constraints, such as flipping in the metal nut category and misplacement in the transistor category. We will consider introducing a memory module to alleviate this problem. On the other hand, the experimental images in this study are mostly well center-aligned, so it is necessary to investigate the robustness of our method on other non-aligned data. In addition, we plan to deploy our algorithm on edge devices for quantization acceleration in future work, making it more convenient for practical applications.

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